Design of an AI Movement Analysis System for Assessing Movement Predictability in MMA Fighters using Pose Estimation

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The Problem & Client / User Context

The Problem

- ***The Problem:*** MMA involves rapid, complex movements: striking, defense, footwork, takedowns, grappling (Zhang, 2025).
- ***The Norm:*** Manual observation is slow and inconsistent (Xu & Nie, 2025)
- ***The Existing Solution:*** *Pose estimation turns body movement into measurable data (Pinheiro et al., 2022; Uçar et al., 2025)*
- ***The Gap:*** Existing work classifies single actions or finds sequences only after manual coding (Kumar et al., 2025; Stefański et al., 2024; Menescardi et al., 2019). None assess recurring patterns and predictability from pose data.

Client / User Context

- **Client:** Underground Battle Mixed Martial Arts (UGB MMA) (headed by Ferdinand Munsayac)
- **Users:** MMA fighters and coaches
- **Purpose:** Support UGB MMA in analyzing fighter movements, particularly recurring movement patterns and tendencies that may contribute to predictability, through a structured, data-based approach.

Proposed Project And Objectives

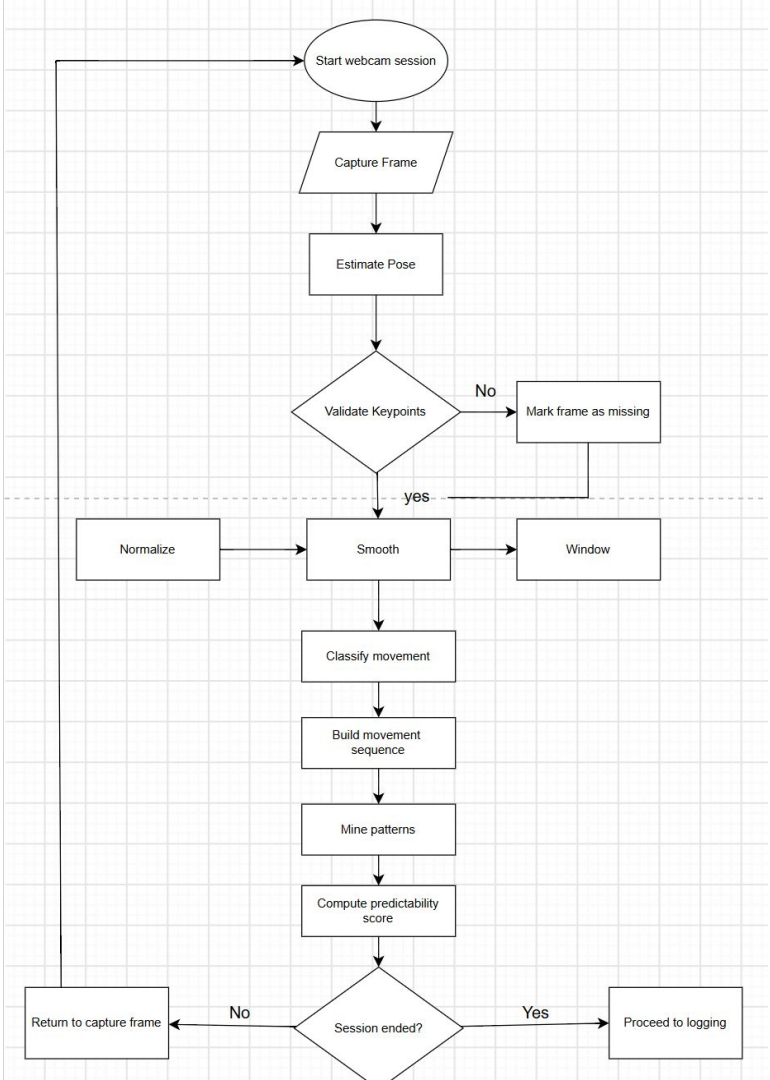
General Objective:

To design and develop a real-time system that detects MMA movements, finds recurring patterns, and assesses the predictability of a single fighter.

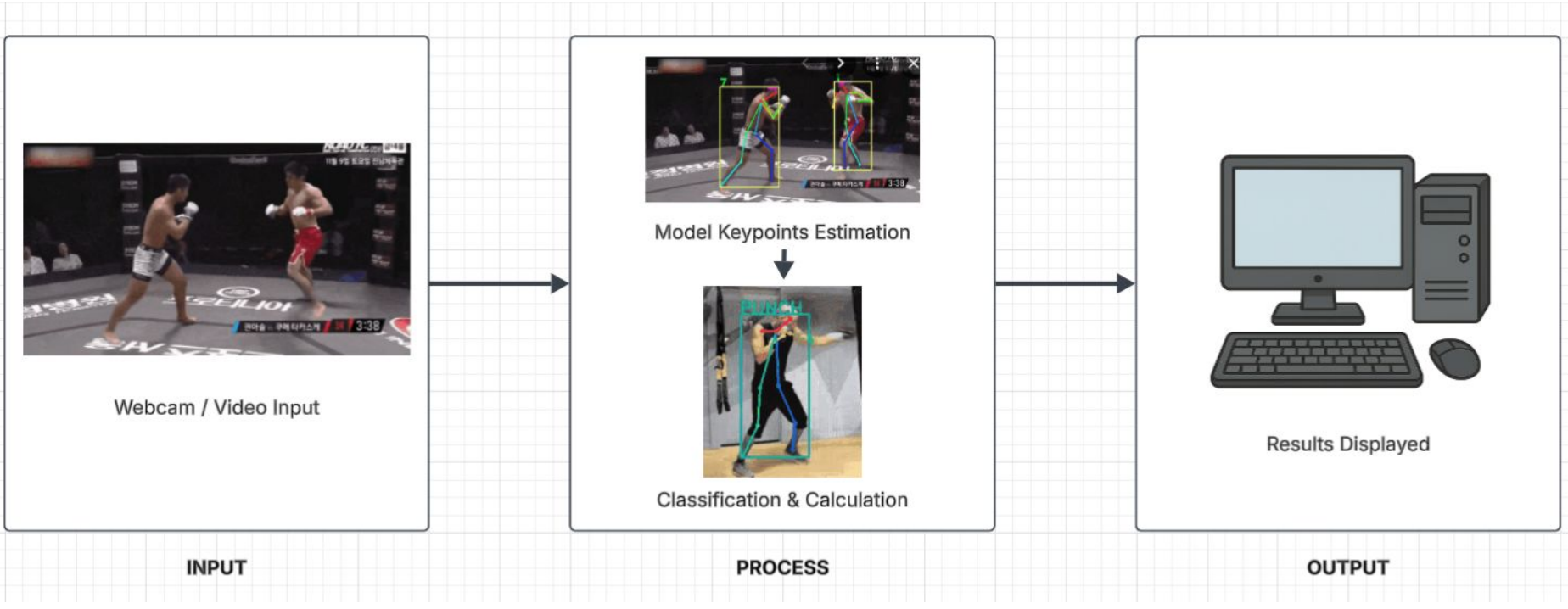
Specific Objectives:

- Track 17 COCO keypoints from webcam in real time
- Classify 5 movements (punch, kick, block, dodge, footwork) at ≥80% accuracy
- Report ≥3 recurring patterns
- Output a 0–100% predictability score, displayed in real time

1. Test and evaluate the design alternatives based on inference time, throughput, peak RAM usage, CPU usage, and storage requirements. Compare the designs using defined constraints and trade-off analysis to determine the most suitable design for the proposed system.



System Design



Scope And Delimitation

Scope

- One MMA Fighter per Session
- Uses webcam to capture a single fighter and processes movement in real time.
- Tracks 17 selected body keypoints; classifies punches, kicks, blocks, dodges, and footwork.

Delimitation

- One fighter at a time
- No specialized webcam / motion capturing device (wearable device)
- Uses pre-trained models and will not develop a new model from scratch (benchmarking only).
- Testing conditions might affect the performance of the system such as lighting, camera angle, blurring, and visibility.
- The system also delimits the prediction of the opponent’s next action, as it only focuses on the main fighter.

Constraints & Metrics, and Engineering Standards

Constraint	Metric
Response time	inference time (ms)
Real-time performance	throughput, images/s
Resource Efficiency	Peak RAM (MB)
Resource Efficiency	CPU usage (% of one core)
Storage Requirement	Storane (MB)

Engineering Standards
ISO/IEC 25010:2023 (Software Product Quality)
ISO/IEC 25059:2023 (Quality Model for AI Systems)
ISO/IEC 25012:2008 (Data Quality Model)
ISO/IEC 27001:2022 (Information Security Management)
ISO 8601-1:2019 (Date and Time Representation)

Functional Design Requirements

Client Requirement	Functional Design Requirement
The system can detect and classify basic MMA movement types.	The system must identify at least 5 predefined MMA movement types, namely punches, kicks, blocks, dodges, and footwork, with at least 80% classification accuracy from recorded test videos.
The system can detect and track the body movement of a single MMA fighter in real time.	The system must track 17 selected body keypoints (the COCO keypoint set; Lin et al., 2014) of one MMA fighter from webcam input during real-time operation.
The system can identify recurring movement patterns and tendencies of MMA fighters.	The system must identify and report at least 3 recurring movement patterns based on the frequency of detected movement sequences.
The system can provide an assessment of a fighter's movement predictability.	The system must generate a movement predictability score from 0% to 100% based on the recurrence and frequency of detected movement patterns.

SDG Alignment



UN SDG 3 (Good Health and Well Being):

Regular physical activity and sports participation contribute to physical and mental health (World Health Organization, 2024). By supporting fighters and coaches with data-driven movement analysis, this project encourages more informed and systematic training practices in support of this goal.

Prior Art / Existing Solutions

Design	Pose-Keypoint Tracking	Movement Classification	Movement Sequence Identification	Movement Predictability Score	Real-Time Operation
Prior Art A: Stefański et al. (2024)		X			X
Prior Art B: Kumar et al. (2025), BoxingVI	X	X			X
Prior Art C: Menescardi et al. (2019)			X		
PROPOSED PROJECT	**X**	**X**	**X**	**X**	**X**

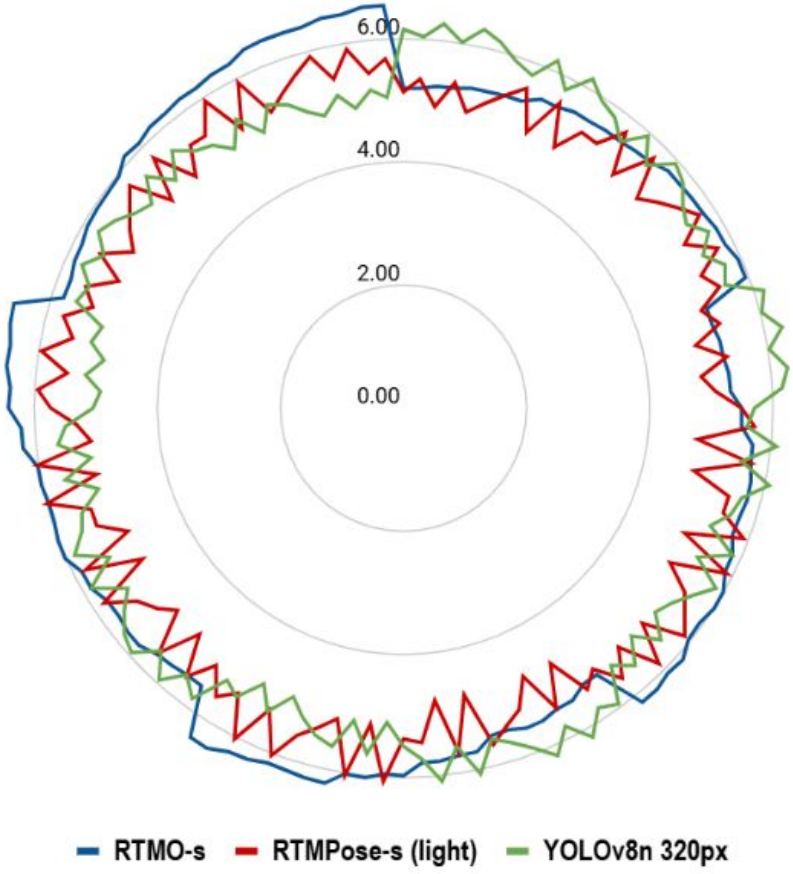
Benchmarking (Raw and Normalized Results)

TABLE 1. RAW RESULTS						
Constraint	**Preference**	**Importance (0-10)**	**Importance (%)**	**Design A: RTMO-s**	**Design B: RTMPose-s (light)**	**Design C: YOLOv8n 320px**
Inference Time (ms)	MIN	10	25.00%	85.34	62.79	19.59
Throughput (images/s)	MAX	9	22.50%	11.53	15.84	50.76
Peak RAM (MB)	MIN	8	20.00%	181.47	116.44	336.20
CPU Usage (%)	MIN	7	17.50%	588.84	1,046.92	791.29
Storage (MB)	MIN	6	15.00%	92.14	94.69	859.94

TABLE 2. NORMALIZED RESULTS AND WEIGHTED SCORE						
Constraint	**Preference**	**Importance (0-10)**	**Importance (%)**	**Design A: RTMO-s**	**Design B: RTMPose-s (light)**	**Design C: YOLOv8n 320px**
Inference Time (ms)	MIN	10	25.00%	1.00	4.09	10.00
Throughput (images/s)	MAX	9	22.50%	1.00	1.99	10.00
Peak RAM (MB)	MIN	8	20.00%	7.34	10.00	1.00
CPU Usage (%)	MIN	7	17.50%	10.00	1.00	6.02
Storage (MB)	MIN	6	15.00%	10.00	9.97	1.00
Weighted score (max 10)		**40**	**100.00%**	**5.19**	**5.14**	**6.15**
Rank				**2**	**3**	**1**

Sensitivity Analysis

	SENSITIVITY
Design	**Wins**
Design A: RTMO-s	70
Design B: RTMPose-s (light)	12
Design C: YOLOv8n 320px	38
Total	**120**



Team Composition

Escalona, Akira Destiny Ashley F.

Team Leader

Lachica, Julz Angelo Matthew L.

Member

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